

Course 5063C

5 Credits: 4

Program Elective

Source-grounded

Long staple spinning Technology

□ To enable the students to understand the method of yarn production from long staple

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 5063C is the authoritative source for course information, objectives, course outcomes, CO-PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 5063C.pdf from the uploaded official SITTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Textile Technology
Course code	5063C
Course title	Long staple spinning Technology
Semester	5 Credits: 4
Course category	Program Elective

Item	Official information
Periods per week	4 (L:4, T:0, P:0) Periods per semester: 60
Periods per semester	60
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

16 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- To enable the students to understand the method of yarn production from long staple fibres like wool and flax using long staple spinning technology.

Course outcomes

CO	Official outcome	Study level
CO1	15 Understanding carding of long staple fibers. Describe the principle and working of drawing	See official syllabus
CO2	10 Understanding machine suitable for long staple fibers Discuss the wool combing operation , worsted	See official syllabus
CO3	top finishing and roving formation process 18 Applying employed in Long staple spinning industry Illustrate the different methods of Long staple Yarn spinning and summarize the yarn	See official syllabus
CO4	15 Applying manufacturing process. Flow chart for long staple fibres.	See official syllabus

CO-PO mapping evidence

Row	Extracted official mapping
CO1	CO1 3 2
CO2	CO2 3 2
CO3	CO3 3 2
CO4	CO4 3 2

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	fibers.	4	As specified
Module 2	staple fibers	4	As specified
Module 3	formation process employed in Long staple spinning industry	4	As specified
Module 4	summarize the yarn manufacturing process flow chart for long staple	4	As specified

MODULE 1

fibers.

This module is presented from the official syllabus scope for Long staple spinning Technology. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: fibers.
- Official duration: As specified

- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

the method of removing impurities in the 3 Remembering wool fibre. Discuss the methods adopted to process raw

the method of removing impurities in the 3 Remembering wool fibre. Discuss the methods adopted to process raw is an official topic in Module 1 of Long staple spinning Technology. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify the method of removing impurities in the 3 Remembering wool fibre. Discuss the methods adopted to process raw using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, the method of removing impurities in the 3 remembering wool fibre. discuss the methods adopted to process raw should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What the method of removing impurities in the 3 Remembering wool fibre. Discuss the methods adopted to process raw means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

[illegible]

flax & jute and Automatic fibre blending 4 Understanding for the woollen system Describe the working principle of carding machine and details of different type of

flax & jute and Automatic fibre blending 4 Understanding for the woollen system Describe the working principle of carding machine and details of different type of is an official topic in Module 1 of Long staple spinning Technology. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify flax & jute and Automatic fibre blending 4 Understanding for the woollen system Describe the working principle of carding machine and details of different type of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, flax & jute and automatic fibre blending 4 understanding for the woollen system describe the working principle of carding machine and details of different type of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What flax & jute and Automatic fibre blending 4 Understanding for the woollen system Describe the working principle of carding machine and details of different type of means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- flax & jute and Automatic fibre blending 4 Understanding for the woollen system Describe the working principle of carding machine and details of different type of **പരപ്പാലി** definition/meaning **പരപ്പാലി**. **പരപ്പാലി** **പരപ്പാലി** **പരപ്പാലി**, steps, application **പരപ്പാലി** **പരപ്പാലി** **പരപ്പാലി**. Technical terms English- **പരപ്പാലി** **പരപ്പാലി**.

4 carding machine-worsted carding, semi - Understanding worsted carding, woollen carding Illustrate the working of Flax carding and

4 carding machine-worsted carding, semi - Understanding worsted carding, woollen carding Illustrate the working of Flax carding and is an official topic in Module 1 of Long staple spinning Technology. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 carding machine-worsted carding, semi - Understanding worsted carding, woollen carding Illustrate the working of Flax carding and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 carding machine-worsted carding, semi - understanding worsted carding, woollen carding illustrate the working of flax carding and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 carding machine-worsted carding, semi - Understanding worsted carding, woollen carding Illustrate the working of Flax carding and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a

diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-4 carding machine-worsted carding, semi - Understanding worsted carding, woollen carding Illustrate the working of Flax carding and definition/meaning . steps, application . Technical terms English- .

4 Understanding jute carding. OPENING, CLEANING AND CARDING

Introduction to long staple fibres - Raw wool processing - Impurities in the wool fibre and their removal - Wool opening and shaking - Wool scouring - Raw Flax Processing - Raw jute processing - Automatic fibre blending for the woollen system - Objectives of carding- working principle and details of different type of carding machine - worsted carding, semi - worsted carding, woollen carding, flax carding and jute carding Describe the principle and working of drawing machine suitable for long

4 Understanding jute carding. OPENING, CLEANING AND CARDING Introduction to long staple fibres - Raw wool processing - Impurities in the wool fibre and their removal - Wool opening and shaking - Wool scouring - Raw Flax Processing - Raw jute processing - Automatic fibre blending for the woollen system - Objectives of carding- working principle and details of different type of carding machine - worsted carding, semi - worsted carding, woollen carding, flax carding and jute carding Describe the principle and working of drawing machine suitable for long is an official topic in Module 1 of Long staple spinning Technology. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding jute carding. OPENING, CLEANING AND CARDING Introduction to long staple fibres - Raw wool processing - Impurities in the wool fibre and their removal - Wool opening and shaking - Wool scouring - Raw Flax Processing - Raw jute processing - Automatic fibre blending for the woollen system - Objectives of carding- working principle and details of different type of carding machine - worsted carding, semi - worsted carding, woollen carding, flax carding and jute carding Describe the principle and working of drawing machine suitable for long using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding jute carding. opening, cleaning and carding introduction to long staple fibres - raw wool processing - impurities in the wool fibre and their removal - wool opening and shaking - wool scouring - raw flax processing - raw jute processing - automatic fibre blending for the woollen system - objectives of carding- working principle and details of different type of carding machine - worsted carding, semi - worsted carding, woollen carding, flax carding and jute carding describe the principle and working of drawing machine suitable for long should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding jute carding. OPENING, CLEANING AND CARDING Introduction to long staple fibres - Raw wool processing - Impurities in the wool fibre and their removal - Wool opening and shaking - Wool scouring - Raw Flax Processing - Raw jute processing - Automatic fibre blending for the woollen system - Objectives of carding- working principle and details of different type of carding machine - worsted carding, semi - worsted carding, woollen carding, flax carding and jute carding Describe the principle and working of drawing machine suitable for long means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- 4 Understanding jute carding. OPENING, CLEANING AND CARDING Introduction to long staple fibres - Raw wool processing - Impurities in the wool fibre and their removal - Wool opening and shaking - Wool scouring - Raw Flax Processing - Raw jute processing - Automatic fibre blending for the woollen system - Objectives of carding- working principle and details of different type of carding machine - worsted carding, semi - worsted carding, woollen carding, flax carding and jute carding Describe the principle and working of drawing machine suitable for long
 definition/meaning
 .
 , steps, application
 .
 . Technical terms English-
 .

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

fibers.

M1.01	Explain the basics of Long staple fibres and the method of removing impurities in the wool fibre.	3
Remembering		
M1.02	Discuss the methods adopted to process raw flax & jute and Automatic fibre blending for the woollen system	4
Understanding		
M1.03	Describe the working principle of carding machine and details of different type of carding machine-worsted carding, semi – worsted carding, woollen carding	4
Understanding		
M1.04	Illustrate the working of Flax carding and jute carding.	4
Understanding		
Contents:		
OPENING, CLEANING AND CARDING		
Introduction to long staple fibres - Raw wool processing - Impurities in the wool fibre and their removal - Wool opening and shaking - Wool scouring - Raw Flax Processing - Raw jute processing - Automatic fibre blending for the woollen system - Objectives of		

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

staple fibers

This module is presented from the official syllabus scope for Long staple spinning Technology. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: staple fibers
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

2 Remembering of long-staple drafting Explain the methods of fibre control in long

2 Remembering of long-staple drafting Explain the methods of fibre control in long is an official topic in Module 2 of Long staple spinning Technology. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Remembering of long-staple drafting Explain the methods of fibre control in long using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 remembering of long-staple drafting explain the methods of fibre control in long should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Remembering of long-staple drafting Explain the methods of fibre control in long means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-പാഠ്യ 2 Remembering of long-staple drafting Explain the methods of fibre control in long പാഠ്യപാഠ്യപാഠ്യപാഠ്യ definition/meaning പാഠ്യപാഠ്യപാഠ്യപാഠ്യ. പാഠ്യപാഠ്യ പാഠ്യപാഠ്യ പാഠ്യപാഠ്യ, steps, application പാഠ്യപാഠ്യ പാഠ്യപാഠ്യ പാഠ്യപാഠ്യ. Technical terms English-പാഠ്യ പാഠ്യ പാഠ്യപാഠ്യപാഠ്യ.

3 Understanding staple drafting zone Discuss the working of worsted, semi

3 Understanding staple drafting zone Discuss the working of worsted, semi is an official topic in Module 2 of Long staple spinning Technology. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding staple drafting zone Discuss the working of worsted, semi using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding staple drafting zone discuss the working of worsted, semi should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding staple drafting zone Discuss the working of worsted, semi means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 2-3 Understanding staple drafting zone Discuss the working of worsted, semi definition/meaning processes. processes, steps, application processes. Technical terms English- processes.

2 Understanding worsted drawing Describe the working of jute and flax

2 Understanding worsted drawing Describe the working of jute and flax is an official topic in Module 2 of Long staple spinning Technology. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding worsted drawing Describe the working of jute and flax using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding worsted drawing describe the working of jute and flax should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding worsted drawing Describe the working of jute and flax means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 2- 2 Understanding worsted drawing Describe the working of jute and flax തുളസി തുളസി definition/meaning തുളസി തുളസി. തുളസി തുളസി തുളസി, steps, application തുളസി തുളസി തുളസി. Technical terms English- തുളസി തുളസി തുളസി.

3 Understanding drawing DRAWING PROCESS Objectives of Gilling process - preparative - Principle of long - staple drafting - fibre control in long staple drafting zone - Pin control, Ambler draft - sliver doubling - Description of worsted, semi worsted, jute and flax drawing - Applications of Autolevelling Discuss the wool combing operation , worsted top finishing and roving

3 Understanding drawing DRAWING PROCESS Objectives of Gilling process - preparative - Principle of long - staple drafting - fibre control in long staple drafting zone - Pin control, Ambler draft - sliver doubling - Description of worsted, semi worsted, jute and flax drawing - Applications of Autolevelling Discuss the wool combing operation , worsted top finishing and roving is an official topic in Module 2 of Long staple spinning Technology. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding drawing DRAWING PROCESS Objectives of Gilling process - preparative - Principle of long - staple drafting - fibre control in long staple drafting zone - Pin control, Ambler draft - sliver doubling - Description of worsted, semi worsted, jute and flax drawing - Applications of Autolevelling Discuss the wool combing operation , worsted top finishing and roving using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding drawing drawing process objectives of gilling process - preparative - principle of long - staple drafting - fibre control in long staple drafting zone - pin control, ambler draft - sliver doubling - description of worsted, semi worsted, jute and flax drawing - applications of autolevelling discuss the wool combing operation , worsted top finishing and roving should be discussed with attention to safe

operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding drawing DRAWING PROCESS Objectives of Gilling process - preparative - Principle of long - staple drafting - fibre control in long staple drafting zone - Pin control, Ambler draft - sliver doubling - Description of worsted, semi worsted, jute and flax drawing - Applications of Autolevelling Discuss the wool combing operation , worsted top finishing and roving means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-3 Understanding drawing DRAWING PROCESS
Objectives of Gilling process - preparative - Principle of long - staple drafting - fibre control in long staple drafting zone - Pin control, Ambler draft - sliver doubling - Description of worsted, semi worsted, jute and flax drawing - Applications of Autolevelling Discuss the wool combing operation , worsted top finishing and roving
രണ്ടാം ഘട്ടം definition/meaning രണ്ടാം ഘട്ടം. രണ്ടാം ഘട്ടം രണ്ടാം ഘട്ടം, steps, application രണ്ടാം ഘട്ടം രണ്ടാം ഘട്ടം. Technical terms English- രണ്ടാം ഘട്ടം രണ്ടാം ഘട്ടം.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

staple fibers		
	List the Objectives of drawing and Principle	
M2.01		2
Remembering		
	of long-staple drafting	
	Explain the methods of fibre control in long	
M2.02		3
Understanding		
	staple drafting zone	
	Discuss the working of worsted, semi	
M2.03		2
Understanding		
	worsted drawing	
	Describe the working of jute and flax	
M2.04		3
Understanding		
	drawing	
	Series Test - I	1
Contents:		
DRAWING PROCESS		
Objectives of Gilling process - preparative - Principle of long - staple drafting		
- fibre		
control in long staple drafting zone - Pin control, Ambler draft - sliver		
doubling -		
Description of worsted, semi worsted, jute and flax drawing - Applications of		

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

formation process employed in Long staple spinning industry

This module is presented from the official syllabus scope for Long staple spinning Technology. Use the topic cards for learning and the source transcript for exact

coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: formation process employed in Long staple spinning industry
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

and the comber preparatory processes 4 Remembering involved. Describe the general features of a rectilinear

and the comber preparatory processes 4 Remembering involved. Describe the general features of a rectilinear is an official topic in Module 3 of Long staple spinning Technology. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify and the comber preparatory processes 4 Remembering involved. Describe the general features of a rectilinear using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, and the comber preparatory processes 4 remembering involved. describe the general features of a rectilinear should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What and the comber preparatory processes 4 Remembering involved. Describe the general features of a rectilinear means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-പരമ്പര and the combinatorial processes 4 Remembering involved. Describe the general features of a rectilinear പരമ്പര definition/meaning പരമ്പര. പരമ്പര പരമ്പര പരമ്പര, steps, application പരമ്പര പരമ്പര പരമ്പര. Technical terms English-പരമ്പര പരമ്പര പരമ്പര.

wool combing machine and the wool 4 Understanding combing cycle sequence of operations Explain the principle of Twist insertion in

wool combing machine and the wool 4 Understanding combing cycle sequence of operations Explain the principle of Twist insertion in is an official topic in Module 3 of Long staple spinning Technology. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify wool combing machine and the wool 4 Understanding combing cycle sequence of operations Explain the principle of Twist insertion in using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, wool combing machine and the wool 4 understanding combing cycle sequence of operations explain the principle of twist insertion in should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What wool combing machine and the wool 4 Understanding combing cycle sequence of operations Explain the principle of Twist insertion in means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a

diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-പൂർണ്ണ വൂൾ കമ്പിംഗ് മെഷീൻ and the wool 4 Understanding combing cycle sequence of operations Explain the principle of Twist insertion in പൂർണ്ണ വൂൾ കമ്പിംഗ് മെഷീൻ definition/meaning പൂർണ്ണ വൂൾ കമ്പിംഗ് മെഷീൻ. പൂർണ്ണ വൂൾ കമ്പിംഗ് മെഷീൻ, steps, application പൂർണ്ണ വൂൾ കമ്പിംഗ് മെഷീൻ പൂർണ്ണ വൂൾ കമ്പിംഗ് മെഷീൻ. Technical terms English-പൂർണ്ണ വൂൾ കമ്പിംഗ് മെഷീൻ.

roving frame and the concept of Flyer lead 5 Understanding delivery Illustrate the operating principle of roving

roving frame and the concept of Flyer lead 5 Understanding delivery Illustrate the operating principle of roving is an official topic in Module 3 of Long staple spinning Technology. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify roving frame and the concept of Flyer lead 5 Understanding delivery Illustrate the operating principle of roving using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, roving frame and the concept of flyer lead 5 understanding delivery illustrate the operating principle of roving should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What roving frame and the concept of Flyer lead 5 Understanding delivery Illustrate the operating principle of roving means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-റോവിംഗ് ഫ്രെയിം and the concept of Flyer lead 5 Understanding delivery Illustrate the operating principle of roving റോവിംഗ് ഫ്രെയിം definition/meaning റോവിംഗ് ഫ്രെയിം. റോവിംഗ് ഫ്രെയിം റോവിംഗ്, steps, application റോവിംഗ് ഫ്രെയിം റോവിംഗ് ഫ്രെയിം. Technical terms English-റോവിംഗ് ഫ്രെയിം റോവിംഗ് ഫ്രെയിം.

5 Understanding machine. COMBING AND ROVING FRAME Objective of combing - basic principles of wool combing - The rectilinear wool comb Preparatory processes - General features of a rectilinear wool comb - wool combing cycle of operations - worsted top finishing - Finisher gilling and packaging - Re - combing Expressing yield and noil production in combing - Tear ratio, Romaine(%) and Noil (%). Twist insertion in roving frame - Flyer lead delivery - Operating principle of roving machine. Illustrate the different methods of Long staple Yarn spinning and

5 Understanding machine. COMBING AND ROVING FRAME Objective of combing - basic principles of wool combing - The rectilinear wool comb Preparatory processes - General features of a rectilinear wool comb - wool combing cycle of operations - worsted top finishing - Finisher gilling and packaging - Re - combing Expressing yield and noil production in combing - Tear ratio, Romaine(%) and Noil (%). Twist insertion in roving frame - Flyer lead delivery - Operating principle of roving machine. Illustrate the different methods of Long staple Yarn spinning and is an official topic in Module 3 of Long staple spinning Technology. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 5 Understanding machine. COMBING AND ROVING FRAME Objective of combing - basic principles of wool combing - The rectilinear wool comb Preparatory processes - General features of a rectilinear wool comb - wool combing cycle of operations - worsted top finishing - Finisher gilling and packaging - Re - combing Expressing yield and noil production in combing - Tear ratio, Romaine(%) and Noil (%). Twist insertion in roving frame - Flyer lead delivery - Operating principle of roving machine. Illustrate the different methods of Long staple Yarn spinning and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 5 understanding machine. combing and roving frame objective of combing - basic principles of wool combing - the rectilinear wool comb preparatory processes - general features of a rectilinear wool comb - wool combing cycle of operations - worsted top finishing - finisher gilling and packaging - re - combing expressing yield and noil production in combing - tear ratio, romaine(%) and noil (%). twist insertion in roving frame - flyer lead delivery - operating principle of roving machine. illustrate the different methods of long staple yarn spinning and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 5 Understanding machine. COMBING AND ROVING FRAME Objective of combing - basic principles of wool combing - The rectilinear wool comb Preparatory processes - General features of a rectilinear wool comb - wool combing cycle of operations - worsted top finishing - Finisher gilling and packaging - Re - combing Expressing yield and noil production in combing - Tear ratio, Romaine(%) and Noil (%). Twist insertion in roving frame - Flyer lead delivery - Operating principle of roving machine. Illustrate the different methods of Long staple Yarn spinning and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- 5 Understanding machine. COMBING AND ROVING FRAME Objective of combing - basic principles of wool combing - The rectilinear wool comb Preparatory processes - General features of a rectilinear wool comb - wool combing cycle of operations - worsted top finishing - Finisher gilling and packaging - Re - combing Expressing yield and noil production in combing - Tear ratio, Romaine(%) and Noil (%). Twist insertion in roving frame - Flyer lead delivery - Operating principle of roving machine. Illustrate the different methods of Long staple Yarn spinning and definition/meaning. steps, application. Technical terms English-.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

formation process employed in Long staple spinning industry

M3.01	Discuss the basic principles of wool combing and the comber preparatory processes	4
Remembering	involved.	

M3.02	Describe the general features of a rectilinear wool combing machine and the wool	4
Understanding	combing cycle sequence of operations	

M3.03	Explain the principle of Twist insertion in roving frame and the concept of Flyer lead	5
Understanding	delivery	

M3.04	Illustrate the operating principle of roving	5
Understanding	machine.	

Contents:

COMBING AND ROVING FRAME

Objective of combing - basic principles of wool combing - The rectilinear wool comb

Preparatory processes - General features of a rectilinear wool comb - wool combing cycle

of operations - worsted top finishing - Finisher gilling and packaging - Re - combing

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

summarize the yarn manufacturing process flow chart for long staple

This module is presented from the official syllabus scope for Long staple spinning Technology. Use the topic cards for learning and the source transcript for exact

coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: summarize the yarn manufacturing process flow chart for long staple
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

wool ring frame and the working 3 Understanding mechanism of woolen Ring spinning frame Explain the working mechanism of Worsted

wool ring frame and the working 3 Understanding mechanism of woolen Ring spinning frame Explain the working mechanism of Worsted is an official topic in Module 4 of Long staple spinning Technology. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify wool ring frame and the working 3 Understanding mechanism of woolen Ring spinning frame Explain the working mechanism of Worsted using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, wool ring frame and the working 3 understanding mechanism of woolen ring spinning frame explain the working mechanism of worsted should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What wool ring frame and the working 3 Understanding mechanism of woolen Ring spinning frame Explain the working mechanism of Worsted means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a

diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-പൂർണ്ണ വൂൾ റിംഗ് ഫ്രെയിം and the working 3
Understanding mechanism of woolen Ring spinning frame Explain the working
mechanism of Worsted പൂർണ്ണ വൂൾ റിംഗ് ഫ്രെയിം definition/meaning പൂർണ്ണ വൂൾ റിംഗ് ഫ്രെയിം.
പൂർണ്ണ വൂൾ റിംഗ് ഫ്രെയിം, steps, application പൂർണ്ണ വൂൾ റിംഗ് ഫ്രെയിം പൂർണ്ണ വൂൾ റിംഗ് ഫ്രെയിം. Technical
terms English-പൂർണ്ണ വൂൾ റിംഗ് ഫ്രെയിം.

spinning frame and Semi worsted spinning 4 Understanding frame Describe the Rings and Traveler used for

spinning frame and Semi worsted spinning 4 Understanding frame Describe the Rings and Traveler used for is an official topic in Module 4 of Long staple spinning Technology. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify spinning frame and Semi worsted spinning 4 Understanding frame Describe the Rings and Traveler used for using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, spinning frame and semi worsted spinning 4 understanding frame describe the rings and traveler used for should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What spinning frame and Semi worsted spinning 4 Understanding frame Describe the Rings and Traveler used for means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-റോട്ടറി spinning frame and Semi worsted spinning 4 Understanding frame Describe the Rings and Traveler used for റോട്ടറി spinning definition/meaning റോട്ടറി spinning. റോട്ടറി spinning റോട്ടറി spinning, steps, application റോട്ടറി spinning റോട്ടറി spinning. Technical terms English-റോട്ടറി spinning റോട്ടറി spinning.

4 Understanding woolen and flax spinning. Illustrate the working of Precision winder

4 Understanding woolen and flax spinning. Illustrate the working of Precision winder is an official topic in Module 4 of Long staple spinning Technology. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding woolen and flax spinning. Illustrate the working of Precision winder using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding woolen and flax spinning. illustrate the working of precision winder should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding woolen and flax spinning. Illustrate the working of Precision winder means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-4 Understanding woolen and flax spinning.

Illustrate the working of Precision winder ഫ്രീഷൻ വൈൻഡർ definition/meaning ഫ്രീഷൻ വൈൻഡർ. ഫ്രീഷൻ വൈൻഡർ steps, application ഫ്രീഷൻ വൈൻഡർ. Technical terms English-ഫ്രീഷൻ വൈൻഡർ.

and illustrate the Process flow chart for 4 Understanding manufacturing woolen, worsted, flax and jute YARN SPINNING AND WINDING Difference between woolen yarn and worsted yarn - Different input material fed to wool ring frame - working mechanism of woollen Ring spinning frame - Drafting and the false twist device - Worsted spinning frame - Semi worsted spinning frame - Double zone drafting system - Lubricated conical rings - J shaped travelers - Flax spinning Ring and Traveller - self - twist spinning system - Yarn winding - Precision winders Process flow chart for manufacturing woollen, worsted, flax and jute Text / Reference: T/R Book Title/Author T1 Oxtoby E., "Spun Yarn Technology", Butterworths, London, Richards RTD and Sykes A B," Woollen Yarn Manufacture", The Textile R2 Institute, Manchester 1994. Sharp P, "Flax, Tow and Jute Spinning", Abhishek Publications, Chandigarh, R3 1998 Stout H P, "Fibre and Yarn Quality in Jute Spinning", The Textile Institute, R4 Manchester, 1988. R5 Chernysheva N, "Wool Spinning" Vol 1&2, Mir Publishers, Moscow, 1983. Online resources: Sl.No Website Link 1 <https://woolery.com/modern-flax-hemp-and-jute-spinning-and-twisting.html> 2 <https://www.woolmark.com> 3 <https://www.bracker.ch> 4 <https://www.woolwise.com/wp-content/uploads>

and illustrate the Process flow chart for 4 Understanding manufacturing woolen, worsted, flax and jute YARN SPINNING AND WINDING Difference between woolen yarn and worsted yarn - Different input material fed to wool ring frame - working mechanism of woollen Ring spinning frame - Drafting and the false twist device - Worsted spinning frame - Semi worsted spinning frame - Double zone drafting system - Lubricated conical rings - J shaped travelers - Flax spinning Ring and Traveller - self - twist spinning system - Yarn winding - Precision winders Process flow chart for manufacturing woollen, worsted, flax and jute Text / Reference: T/R Book Title/Author T1 Oxtoby E., "Spun Yarn Technology", Butterworths, London, Richards RTD and Sykes A B," Woollen Yarn Manufacture", The Textile R2 Institute, Manchester 1994. Sharp P, "Flax, Tow and Jute Spinning", Abhishek Publications, Chandigarh, R3 1998 Stout H P, "Fibre and Yarn Quality in Jute Spinning", The Textile Institute, R4 Manchester, 1988. R5 Chernysheva N, "Wool Spinning" Vol 1&2, Mir Publishers, Moscow, 1983. Online resources: Sl.No Website Link 1 <https://woolery.com/modern-flax-hemp-and-jute-spinning-and-twisting.html> 2 <https://www.woolmark.com> 3 <https://www.bracker.ch> 4 <https://www.woolwise.com/wp-content/uploads> is an official topic in Module 4 of Long staple spinning Technology. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify and illustrate the Process flow chart for 4 Understanding manufacturing woolen, worsted, flax and jute YARN SPINNING AND WINDING
Difference between woolen yarn and worsted yarn - Different input material fed to wool ring frame - working mechanism of woollen Ring spinning frame - Drafting and the false twist device - Worsted spinning frame - Semi worsted spinning frame - Double zone drafting system - Lubricated conical rings - J shaped travelers - Flax spinning Ring and Traveller - self - twist spinning system - Yarn winding - Precision winders Process flow chart for manufacturing woollen, worsted, flax and jute Text / Reference: T/R Book Title/Author T1 Oxtoby E., "Spun Yarn Technology", Butterworths, London, Richards RTD and Sykes A B," Woollen Yarn Manufacture", The Textile R2 Institute, Manchester 1994. Sharp P, "Flax, Tow and Jute Spinning", Abhishek Publications, Chandigarh, R3 1998 Stout H P, "Fibre and Yarn Quality in Jute Spinning", The Textile Institute, R4 Manchester, 1988. R5 Chernysheva N, "Wool Spinning" Vol 1&2, Mir Publishers, Moscow, 1983. Online resources: Sl.No Website Link 1 <https://woolery.com/modern-flax-hemp-and-jute-spinning-and-twisting.html> 2 <https://www.woolmark.com> 3 <https://www.bracker.ch> 4 <https://www.woolwise.com/wp-content/uploads> using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, and illustrate the process flow chart for 4 understanding manufacturing woolen, worsted, flax and jute yarn spinning and winding difference between woolen yarn and worsted yarn - different input material fed to wool ring frame - working mechanism of woollen ring spinning frame - drafting and the false twist device - worsted spinning frame - semi worsted spinning frame - double zone drafting system - lubricated conical rings - j shaped travelers - flax spinning ring and traveller - self - twist spinning system - yarn winding - precision winders process flow chart for manufacturing woollen, worsted, flax and jute text / reference: t/r book title/author t1 oxtoby e., "spun yarn technology", butterworths, london, richards rtd and sykes a b," woollen yarn manufacture", the textile r2 institute, manchester 1994. sharp p, "flax, tow and jute spinning", abhishek publications, chandigarh, r3 1998 stout h p, "fibre and yarn quality in jute spinning", the textile institute, r4 manchester, 1988. r5 chernysheva

n, “wool spinning” vol 1&2, mir publishers, moscow, 1983. online resources: sl.no website link 1 <https://woolery.com/modern-flax-hemp-and-jute-spinning-and-twisting.html> 2 <https://www.woolmark.com> 3 <https://www.bracker.ch> 4 <https://www.woolwise.com/wp-content/uploads> should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What and illustrate the Process flow chart for 4 Understanding manufacturing woollen, worsted, flax and jute YARN SPINNING AND WINDING Difference between woollen yarn and worsted yarn - Different input material fed to wool ring frame - working mechanism of woollen Ring spinning frame - Drafting and the false twist device - Worsted spinning frame - Semi worsted spinning frame - Double zone drafting system - Lubricated conical rings - J shaped travelers - Flax spinning Ring and Traveller - self - twist spinning system - Yarn winding - Precision winders Process flow chart for manufacturing woollen, worsted, flax and jute Text / Reference: T/R Book Title/Author T1 Oxtoby E., “Spun Yarn Technology”, Butterworths, London, Richards RTD and Sykes A B,” Woollen Yarn Manufacture”, The Textile R2 Institute, Manchester 1994. Sharp P, “Flax, Tow and Jute Spinning”, Abhishek Publications, Chandigarh, R3 1998 Stout H P, “Fibre and Yarn Quality in Jute Spinning”, The Textile Institute, R4 Manchester, 1988. R5 Chernysheva N, “Wool Spinning” Vol 1&2, Mir Publishers, Moscow, 1983. Online resources: Sl.No Website Link 1 https://woolery.com/modern-flax-hemp-and-jute-spinning-and-twisting.html 2 https://www.woolmark.com 3 https://www.bracker.ch 4 https://www.woolwise.com/wp-content/uploads means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4- and illustrate the Process flow chart for 4 Understanding manufacturing woollen, worsted, flax and jute YARN SPINNING AND WINDING Difference between woollen yarn and worsted yarn - Different input material fed to wool ring frame - working mechanism of woollen Ring spinning frame - Drafting and the false twist device - Worsted spinning frame - Semi worsted spinning frame - Double zone drafting system - Lubricated conical rings - J shaped travelers - Flax spinning Ring and Traveller - self - twist spinning system - Yarn winding - Precision winders Process flow chart for manufacturing woollen, worsted, flax and jute Text / Reference: T/R Book Title/Author T1 Oxtoby E., “Spun Yarn Technology”, Butterworths, London, Richards RTD and Sykes A B,” Woollen Yarn Manufacture”, The Textile R2 Institute, Manchester 1994. Sharp P, “Flax, Tow and Jute Spinning”, Abhishek Publications, Chandigarh, R3 1998 Stout H P, “Fibre and

Yarn Quality in Jute Spinning”, The Textile Institute, R4 Manchester, 1988. R5 Chernysheva N, “Wool Spinning” Vol 1&2, Mir Publishers, Moscow, 1983. Online resources: Sl.No Website Link 1 <https://woolery.com/modern-flax-hemp-and-jute-spinning-and-twisting.html> 2 <https://www.woolmark.com> 3 <https://www.bracker.ch> 4 <https://www.woolwise.com/wp-content/uploads> [wool spinning definition/meaning](#) [wool spinning definition/meaning](#). [wool spinning definition/meaning](#), steps, application [wool spinning definition/meaning](#) [wool spinning definition/meaning](#). Technical terms English-[wool spinning definition/meaning](#) [wool spinning definition/meaning](#).

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

summarize the yarn manufacturing process flow chart for long staple fibres.

M4.01	Discuss the different input material fed to wool ring frame and the working mechanism of woollen Ring spinning frame	3
Understanding	Explain the working mechanism of Worsted spinning frame and Semi worsted spinning frame	4
M4.02	Describe the Rings and Traveler used for woolen and flax spinning.	4
Understanding	Illustrate the working of Precision winder and illustrate the Process flow chart for manufacturing woollen, worsted, flax and jute	4
M4.03	Series Test - II	1
Understanding		

Contents:

YARN SPINNING AND WINDING

Difference between woollen yarn and worsted yarn - Different input material fed to wool

ring frame - working mechanism of woollen Ring spinning frame - Drafting and the false

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

Definition → Principle → Components/steps → Application → Conclusion

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

Aim → Apparatus → Procedure → Observation → Result → Safety

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

The official assessment structure is reproduced from the supplied PDF.

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

Define or explain the method of removing impurities in the 3 Remembering wool fibre. Discuss the methods adopted to process raw. (3 marks)

Answer: Begin with the standard definition or identity of *the method of removing impurities in the 3 Remembering wool fibre. Discuss the methods adopted to process raw*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain flax & jute and Automatic fibre blending 4 Understanding for the woollen system Describe the working principle of carding machine and details of different type of. (3 marks)

Answer: Begin with the standard definition or identity of *flax & jute and Automatic fibre blending 4 Understanding for the woollen system Describe the working principle of carding machine and details of different type of*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 4 carding machine-worsted carding, semi - Understanding worsted carding, woollen carding Illustrate the working of Flax carding and. (3 marks)

Answer: Begin with the standard definition or identity of *4 carding machine-worsted carding, semi - Understanding worsted carding, woollen carding Illustrate the working of Flax carding and*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Define or explain 4 Understanding jute carding. OPENING, CLEANING AND CARDING Introduction to long staple fibres - Raw wool processing - Impurities in the wool fibre and their removal - Wool opening and shaking - Wool scouring - Raw Flax Processing - Raw jute processing - Automatic fibre blending for the woollen system - Objectives of carding- working principle and details of different type of carding machine - worsted carding, semi - worsted carding, woollen carding, flax carding and jute carding Describe the principle and working of drawing machine suitable for long. (3 marks)

Answer: Begin with the standard definition or identity of 4 *Understanding jute carding. OPENING, CLEANING AND CARDING Introduction to long staple fibres - Raw wool processing - Impurities in the wool fibre and their removal - Wool opening and shaking - Wool scouring - Raw Flax Processing - Raw jute processing - Automatic fibre blending for the woollen system - Objectives of carding- working principle and details of different type of carding machine - worsted carding, semi - worsted carding, woollen carding, flax carding and jute carding Describe the principle and working of drawing machine suitable for long.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Module 2

Define or explain 2 Remembering of long-staple drafting Explain the methods of fibre control in long. (3 marks)

Answer: Begin with the standard definition or identity of 2 *Remembering of long-staple drafting Explain the methods of fibre control in long.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: ഓരോന്നിന്റെയും നിർവ്വചനം, പ്രധാനപ്പെട്ടതുകൾ/പ്രവൃത്തിരീതികൾ, പ്രയോഗങ്ങൾ എന്നിവയെക്കുറിച്ചുള്ള വിവരങ്ങൾ.

Define or explain 3 Understanding staple drafting zone Discuss the working of worsted, semi. (3 marks)

Answer: Begin with the standard definition or identity of *3 Understanding staple drafting zone* Discuss the working of *worsted, semi*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: ഓരോന്നിന്റെയും നിർവ്വചനം, പ്രധാനപ്പെട്ടതുകൾ/പ്രവൃത്തിരീതികൾ, പ്രയോഗങ്ങൾ എന്നിവയെക്കുറിച്ചുള്ള വിവരങ്ങൾ.

Define or explain 2 Understanding worsted drawing Describe the working of jute and flax. (3 marks)

Answer: Begin with the standard definition or identity of *2 Understanding worsted drawing* Describe the working of *jute and flax*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: ഓരോന്നിന്റെയും നിർവ്വചനം, പ്രധാനപ്പെട്ടതുകൾ/പ്രവൃത്തിരീതികൾ, പ്രയോഗങ്ങൾ എന്നിവയെക്കുറിച്ചുള്ള വിവരങ്ങൾ.

Define or explain 3 Understanding drawing DRAWING PROCESS Objectives of Gilling process - preparative - Principle of long - staple drafting - fibre control in long staple drafting zone - Pin control, Ambler draft - sliver doubling - Description of worsted, semi worsted, jute and flax drawing - Applications of Autolevelling Discuss the wool combing operation , worsted top finishing and roving. (3 marks)

Answer: Begin with the standard definition or identity of *3 Understanding drawing* DRAWING PROCESS Objectives of Gilling process - preparative - Principle of long - staple drafting - fibre control in long staple drafting zone - Pin control, Ambler draft - sliver doubling - Description of worsted, semi worsted, jute and flax drawing - Applications of Autolevelling Discuss the wool combing operation , worsted top finishing and roving.

State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Module 3

Define or explain and the comber preparatory processes 4 Remembering involved. Describe the general features of a rectilinear. (3 marks)

Answer: Begin with the standard definition or identity of *and the comber preparatory processes 4 Remembering involved. Describe the general features of a rectilinear*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain wool combing machine and the wool 4 Understanding combing cycle sequence of operations Explain the principle of Twist insertion in. (3 marks)

Answer: Begin with the standard definition or identity of *wool combing machine and the wool 4 Understanding combing cycle sequence of operations Explain the principle of Twist insertion in*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain roving frame and the concept of Flyer lead 5 Understanding delivery Illustrate the operating principle of roving. (3 marks)

Answer: Begin with the standard definition or identity of *roving frame and the concept of Flyer lead* 5 Understanding delivery Illustrate the operating principle of roving. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Define or explain 5 Understanding machine. COMBING AND ROVING FRAME
Objective of combing - basic principles of wool combing - The rectilinear wool comb Preparatory processes - General features of a rectilinear wool comb - wool combing cycle of operations - worsted top finishing - Finisher gilling and packaging - Re - combing Expressing yield and noil production in combing - Tear ratio, Romaine(%) and Noil (%). Twist insertion in roving frame - Flyer lead delivery - Operating principle of roving machine. Illustrate the different methods of Long staple Yarn spinning and. (3 marks)

Answer: Begin with the standard definition or identity of 5 Understanding machine. COMBING AND ROVING FRAME Objective of combing - basic principles of wool combing - The rectilinear wool comb Preparatory processes - General features of a rectilinear wool comb - wool combing cycle of operations - worsted top finishing - Finisher gilling and packaging - Re - combing Expressing yield and noil production in combing - Tear ratio, Romaine(%) and Noil (%). Twist insertion in roving frame - Flyer lead delivery - Operating principle of roving machine. Illustrate the different methods of Long staple Yarn spinning and. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Module 4

Define or explain wool ring frame and the working 3 Understanding mechanism of woollen Ring spinning frame Explain the working mechanism of Worsted. (3 marks)

Answer: Begin with the standard definition or identity of *wool ring frame and the working 3 Understanding mechanism of woollen Ring spinning frame Explain the working mechanism of Worsted*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain spinning frame and Semi worsted spinning 4 Understanding frame Describe the Rings and Traveler used for. (3 marks)

Answer: Begin with the standard definition or identity of *spinning frame and Semi worsted spinning 4 Understanding frame Describe the Rings and Traveler used for*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 4 Understanding woolen and flax spinning. Illustrate the working of Precision winder. (3 marks)

Answer: Begin with the standard definition or identity of *4 Understanding woolen and flax spinning. Illustrate the working of Precision winder*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain and illustrate the Process flow chart for 4 Understanding manufacturing woolen, worsted, flax and jute YARN SPINNING AND WINDING Difference between woolen yarn and worsted yarn - Different input material fed to wool ring frame - working mechanism of woollen Ring spinning frame - Drafting and the false twist device - Worsted spinning frame - Semi worsted

spinning frame - Double zone drafting system - Lubricated conical rings - J shaped travelers - Flax spinning Ring and Traveller - self - twist spinning system - Yarn winding - Precision winders Process flow chart for manufacturing woollen, worsted, flax and jute Text / Reference: T/R Book Title/Author T1 Oxtoby E., "Spun Yarn Technology", Butterworths, London, Richards RTD and Sykes A B," Woollen Yarn Manufacture", The Textile R2 Institute, Manchester 1994. Sharp P, "Flax, Tow and Jute Spinning", Abhishek Publications, Chandigarh, R3 1998 Stout H P, "Fibre and Yarn Quality in Jute Spinning", The Textile Institute, R4 Manchester, 1988. R5 Chernysheva N, "Wool Spinning" Vol 1&2, Mir Publishers, Moscow, 1983. Online resources: Sl.No Website Link 1 <https://woolery.com/modern-flax-hemp-and-jute-spinning-and-twisting.html> 2 <https://www.woolmark.com> 3 <https://www.bracker.ch> 4 <https://www.woolwise.com/wp-content/uploads>. (3 marks)

Answer: Begin with the standard definition or identity of and illustrate the Process flow chart for 4 Understanding manufacturing woollen, worsted, flax and jute YARN SPINNING AND WINDING Difference between woollen yarn and worsted yarn - Different input material fed to wool ring frame - working mechanism of woollen Ring spinning frame - Drafting and the false twist device - Worsted spinning frame - Semi worsted spinning frame - Double zone drafting system - Lubricated conical rings - J shaped travelers - Flax spinning Ring and Traveller - self - twist spinning system - Yarn winding - Precision winders Process flow chart for manufacturing woollen, worsted, flax and jute Text / Reference: T/R Book Title/Author T1 Oxtoby E., "Spun Yarn Technology", Butterworths, London, Richards RTD and Sykes A B," Woollen Yarn Manufacture", The Textile R2 Institute, Manchester 1994. Sharp P, "Flax, Tow and Jute Spinning", Abhishek Publications, Chandigarh, R3 1998 Stout H P, "Fibre and Yarn Quality in Jute Spinning", The Textile Institute, R4 Manchester, 1988. R5 Chernysheva N, "Wool Spinning" Vol 1&2, Mir Publishers, Moscow, 1983. Online resources: Sl.No Website Link 1 <https://woolery.com/modern-flax-hemp-and-jute-spinning-and-twisting.html> 2 <https://www.woolmark.com> 3 <https://www.bracker.ch> 4 <https://www.woolwise.com/wp-content/uploads>. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: ഐക്യ പ്രതിബദ്ധതയ്ക്ക് നിർവ്വചനം, പ്രതിബദ്ധതയ്ക്ക് തത്വം/പ്രവൃത്തി, പ്രതിബദ്ധതയ്ക്ക് പ്രയോഗം എന്നിവ ഉൾപ്പെടെയുള്ളവ.

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **the method of removing impurities in the 3 Remembering wool fibre. Discuss the methods adopted to process raw.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **2 Remembering of long-staple drafting Explain the methods of fibre control in long.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **and the comber preparatory processes 4 Remembering involved. Describe the general features of a rectilinear.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 4

1-mark definition: State the meaning of **wool ring frame and the working 3 Understanding mechanism of woolen Ring spinning frame Explain the working mechanism of Worsted.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.

Term	Meaning in this lesson
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl.No Website Link
- <https://woolery.com/modern-flax-hemp-and-jute-spinning-and-twisting.html>
<https://www.woolmark.com> <https://www.bracker.ch> <https://www.woolwise.com/wp-content/uploads>

IN-PAGE SEARCH

Search this lesson

Prepared from the supplied official SITTTTR syllabus PDF for Course 5063C. Verify institutional updates against the current official curriculum.